

Zigbee



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Intro to Zigbee



- Wireless communication protocol based on IEEE 802.15.4
- Short-range, low-data-rate, low-power use
- Designed mainly for Internet of Things (IoT)
- Optimized for energy efficiency, reliability, scalability
- Used in smart homes, industry, healthcare, and energy systems



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Key Features



Range

10–100 m indoors
(extendable with
mesh routing)



Topologies

Supports Star, Tree,
Mesh topologies



Data Rate

Up to 250 kbps
(sufficient for
control & sensors)



Device Types

Coordinator,
Router, End Device



Frequencies

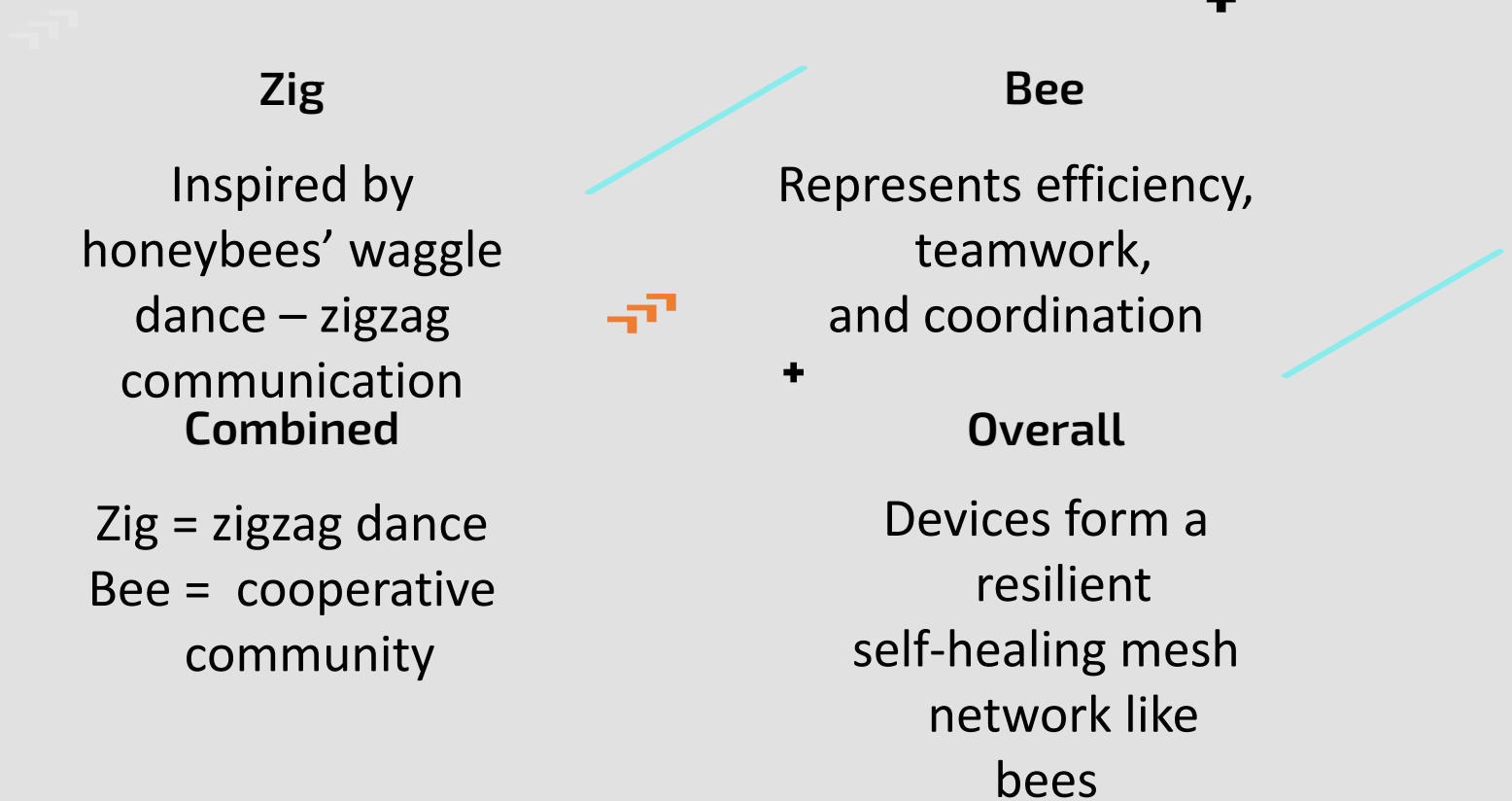
2.4 GHz (global),
868 MHz (EU),
915 MHz (US)



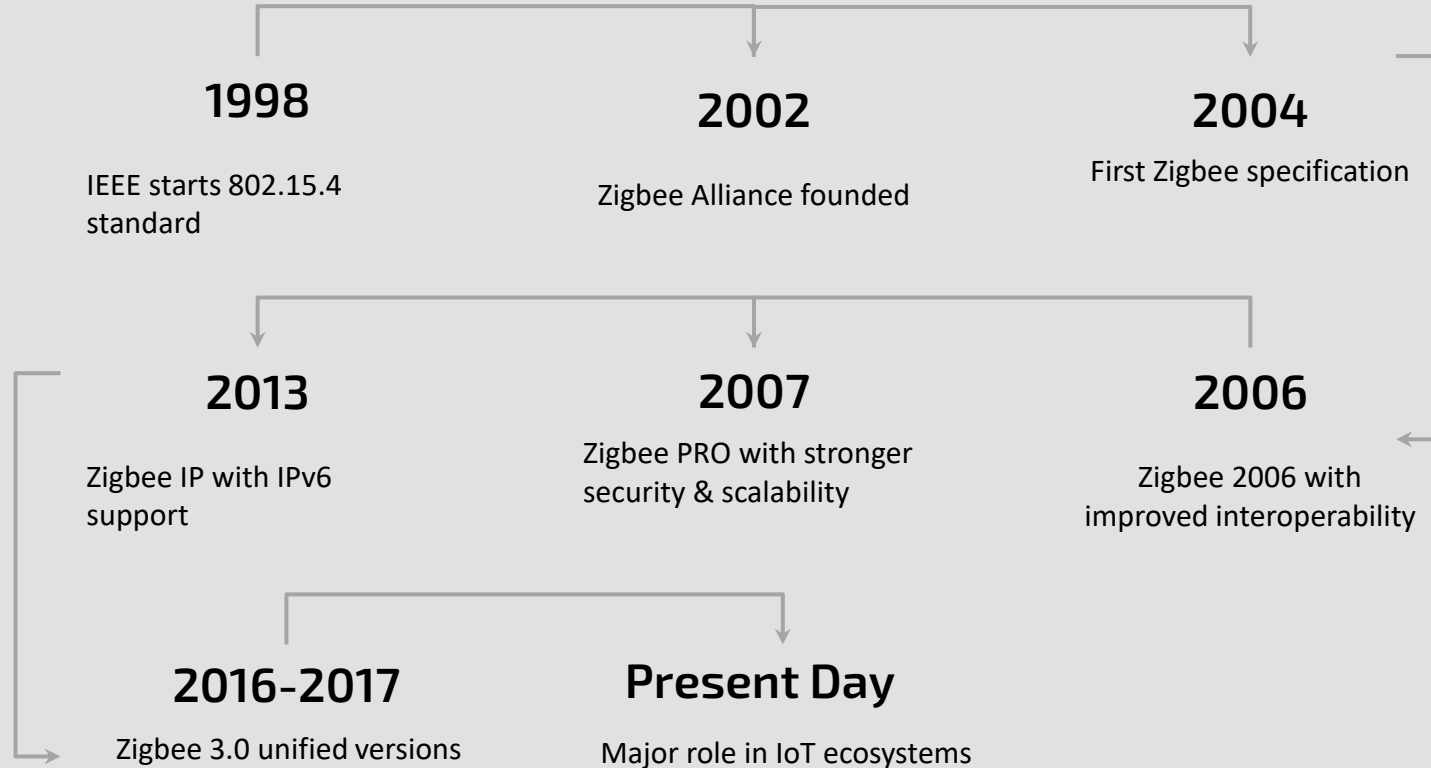
Benefits

Low-cost, highly
scalable, secure
communication

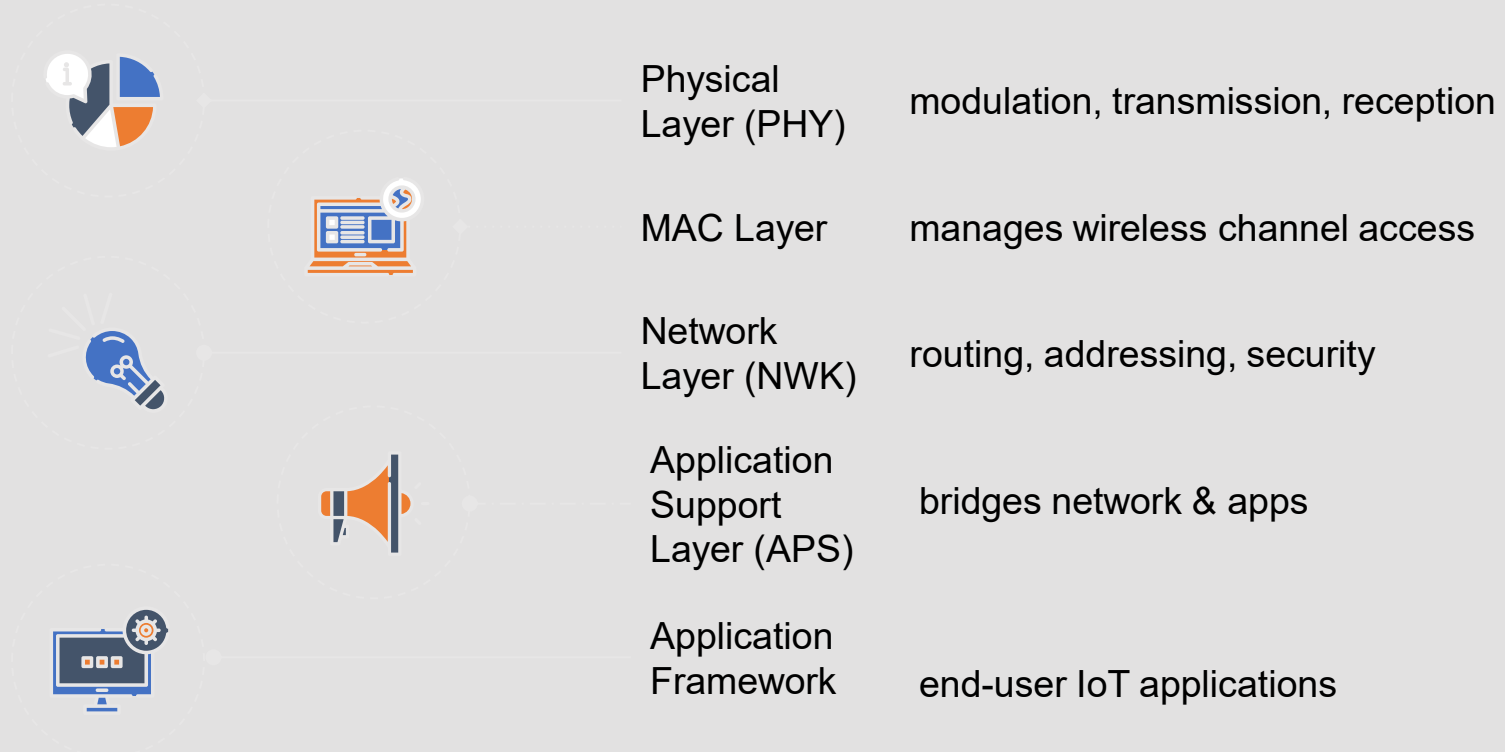
Why Named Zigbee?



Timeline

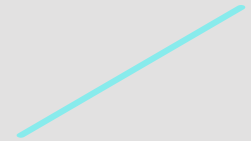


Layers in Zigbee





Topologies



Star

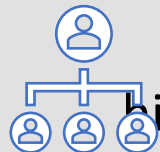


central coordinator as
hub (simple but
single point of
failure)

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Tree



hierarchical parent-
child
connections (scalable)

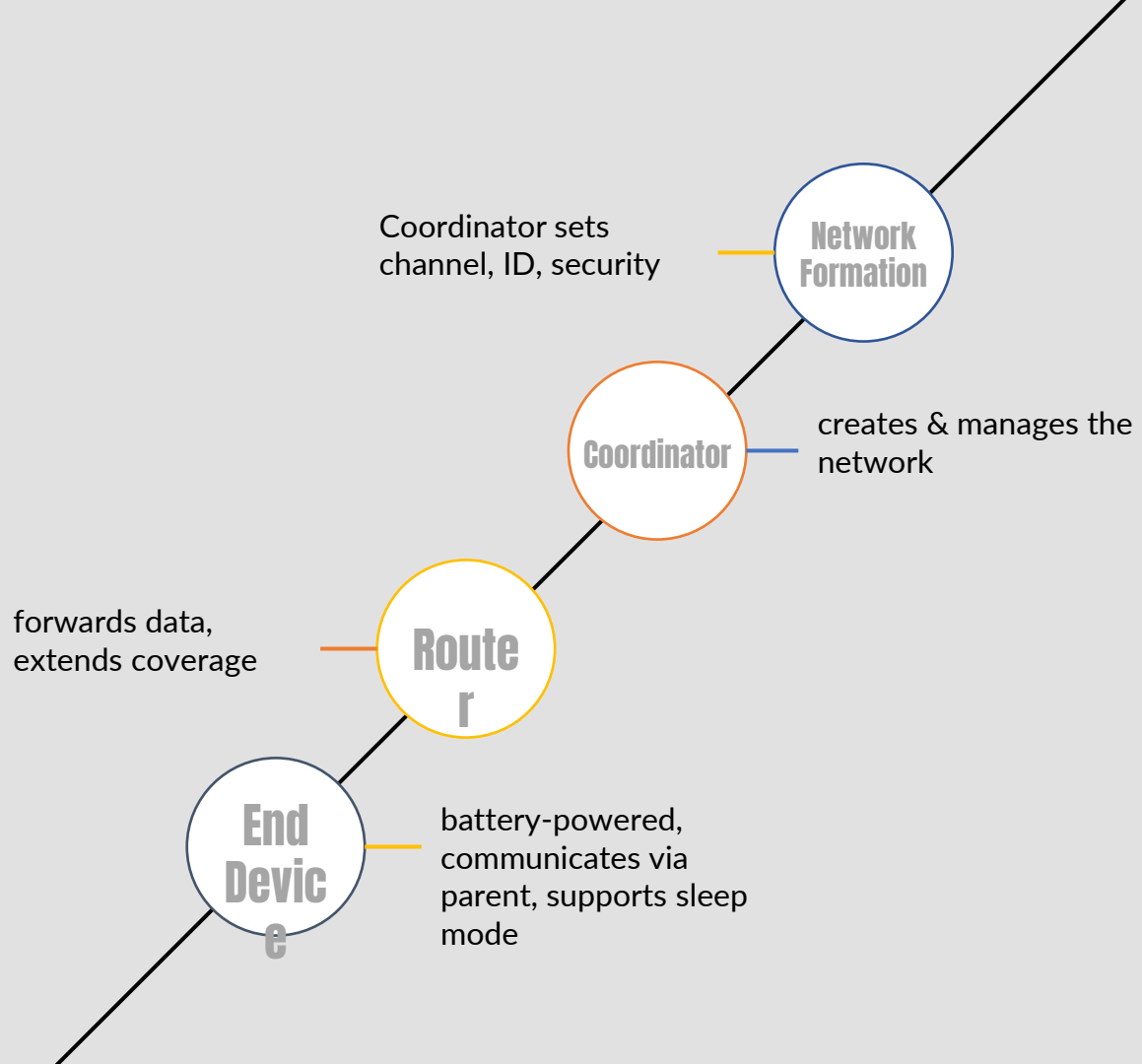


Mesh

devices relay data,
robust and self-healing

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Part 1: Setup & Roles



Part 2: Communication

Low Power Operation

end devices
mostly sleep,
wake to
send/receive



Security

128-bit AES encryption
ensures safe data transfer

Self Healing

If path fails,
alternate routes
used



Routers

enable multi-
hop mesh
routing



Communication

devices send
small data
packets





Ultra-low power consumption →
years of battery life



Highly scalable – supports
thousands of devices



Reliable – self-healing mesh
ensures communication



Secure – AES-128 encryption



Low cost – cheaper than Wi-Fi/Bluetooth

Advantages

Limitations



Low data rate

Maximum speed is 250 kbps, which is much lower than Wi-Fi or Bluetooth. Not suitable for bandwidth-heavy applications like video or large file transfer.

Range

Typically supports 10–100 meters indoors depending on walls and interference. Extended range requires mesh networking, but without enough intermediate devices, coverage is limited.

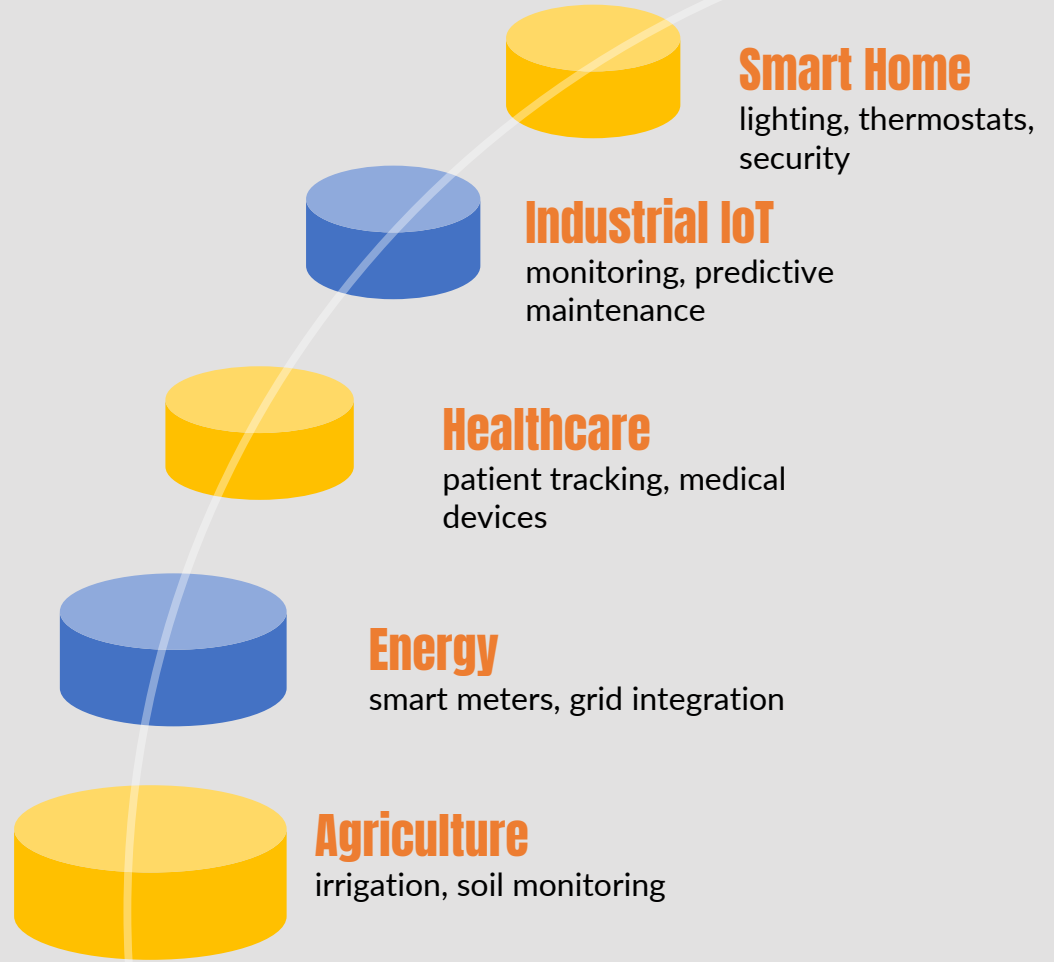
Interference

Operates mostly on the 2.4 GHz ISM band, where Wi-Fi, Bluetooth, and even microwave ovens also operate. Can suffer from congestion and signal overlap in crowded environments.

Latency in Mesh Networks

Multi-hop communication introduces delays. Not ideal for real-time or time-critical applications.

Applications





Thank You